

Nguyen Thanh Minh Duy

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Professional Summary

Recent Master's graduate with 2 years of production software engineering at Zalo and a strong foundation in applied machine learning. Experienced in developing scalable software features, deep learning research pipelines, applied AI experience in multimodal retrieval and RAG-based applications using PyTorch, FAISS, FastAPI, and Docker, with interest in Machine Learning Engineer and AI Engineer roles.

Technical Skills

- AI/ML Research:** PyTorch, PyTorch Geometric (PyG), Generative AI, Multi-modal ML (CLIP/SigLIP), Transformers, GNN, Model Optimization, Anomaly Detection
- Languages & Core:** Python, JavaScript/TypeScript, Data Structures & Algorithms
- ML Frameworks & Tools:** Agentic AI Systems, RAG Systems, FAISS, MCP, Prompt Engineering, Algorithm Implementation, Experimental Design, Data Preprocessing, Model Validation & Evaluation
- Development:** React.js, Electron.js, Node.js, FastAPI, Docker, Git, LangChain, LangGraph

Work Experience

Soonchunhyang University (SCH)
Graduate Research Assistant (Lab: IoT Network)

Asan, South Korea
Feb 2024 – Feb 2026

- Researched and implemented deep learning models in PyTorch, including Transformers and GNNs, for in-vehicle CAN bus intrusion detection under data-scarce, imbalanced, and open-set settings.
- Applied Few-Shot Learning and Open-Set Recognition to detect known and unknown attacks, achieving state-of-the-art multi-class results and up to 95% accuracy on unseen classes.
- Built automated GPU-based training and evaluation pipelines, produced weekly experiment reports with visualizations, and contributed to a Q1 journal publication.

Zalo - VNG Corporation
Software Engineer (React.js/Electron.js)

Ho Chi Minh City, Vietnam
May 2022 – Mar 2024

- Optimized client-side message search using batching and caching, reducing search latency by 30% while maintaining stable UI performance across varied database sizes.
- Shipped 15+ production features for Zalo PC, including a full architectural revamp of the user Contact Card, while maintaining and extending complex legacy codebases.
- Collaborated with Product, Design, and Backend teams to convert requirements into technical solutions; promoted from Fresher to Software Engineer for consistent delivery.

Projects

Bé Đậu (daucv.com) | AI Career SaaS · Founder & Full-Stack Developer

2025 – Present

- Full-Stack Product Ownership:** Built and deployed a localized AI career platform for the Vietnamese market featuring ATS-optimized CV tailoring and real-time voice mock interviews (Web Speech API), using Next.js, FastAPI end-to-end.
- LLM Backend Engineering:** Architected a stateless FastAPI backend integrating OpenAI Structured Outputs (Pydantic) for multi-agent analytical dashboards. Implemented an LLM Waterfall Fallback system (Gemini → Groq → OpenRouter) to ensure 100% uptime and eliminate server costs.
- AI Interview Simulation:** Implemented dynamic interview personas through prompt engineering to simulate HR, technical, and managerial interview scenarios.

Interactive Video Retrieval System | HCMC AI Competition (Finalist)

Aug 2025 - Oct 2025

- Engineered a multi-modal AI (CLIP/SigLIP) retrieval system for text-to-keyframe matching, integrated with FAISS for fast and scalable vector similarity search.
- Processed dense multimedia data by automating shot detection and audio/video frame extraction via ffmpeg and TransNet, optimizing the architecture for large-scale indexing.
- Implemented a responsive React UI for dynamic query refinement and result visualization.

Publications

- Nguyen T. M. Duy**, et al. *Multi-Class Intrusion Detection System for In-Vehicle Networks Using Few-Shot Learning and Convolutional Anomaly Transformer*. **Knowledge-Based Systems** (Q1), Nov 2025.
- Open-Set Recognition with Multi-Objective Graph Neural Network in CAN*. Under review, 2025.

Education & Certifications

Certifications: Generative AI Engineering with LLMs Specialization (IBM)

Mar 2026

Soonchunhyang University (SCH)
Master of Science in Engineering (Fully Funded); GPA: 4.32/4.5

South Korea
March 2024 – Feb 2026

HCM VNU - University of Science
B.Sc. in Software Engineering; GPA: 8.66/10

Ho Chi Minh City, VietNam
Aug 2018 – Feb 2022